NATURAL LANGUAGE PROCESSING

M-DAS

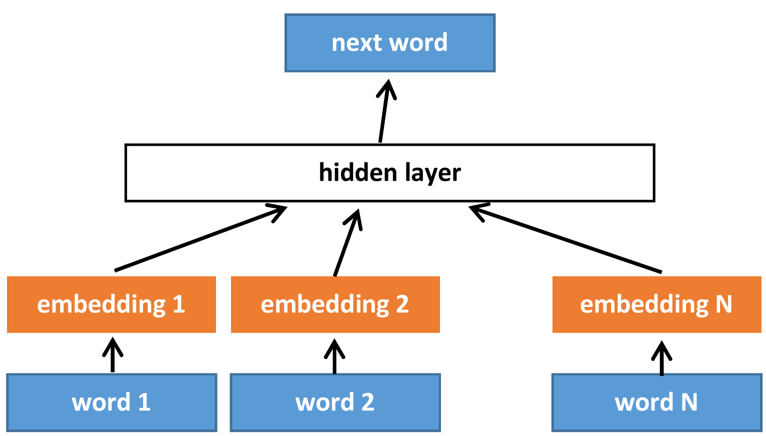
Mini Project 3 – Word Embeddings

I. Given a Khmer text corpus “temples.txt” extracted from 3 Wikipedia pages[1], build a skip-gram model/classifier similar to what you have done for Exercise 4.4 to find representation/embedding of each Khmer word in the corpus. Use the following settings for your implementation:

- The word embedding has a dimension of 50
- For the skip-gram model, use context window $L = \pm 4$ and negative sampling with $k = 2$
- To tokenize word, you can use Khmer nltk[2]
- Your vocabulary should contain meaningful and frequent words so:
    - Words whose frequency is less than 10 are ignored
    - Spaces are considered a stop word and are also ignored

II. Apply dimensionality reduction using PCA and select only the two most important components from the word embedding, and plot each word representation on a 2D graph.

III. Build a neural language model which is an alternative architecture to the n-gram model that you have seen in Lesson 02. The model predicts the next word given the N previous words. This is done by concatenating the word embeddings of N previous words and use them as input of a single hidden layer of size H with a non-linearity (e.g. sigmoid). Finally, a softmax layer is used to make a prediction of the next word. Train your model with $N = 5$ and $H = 512$ (you can also propose your own architecture).



[1] https://km.wikipedia.org/wiki/{ប្រាសាទអង្គរវត្ត, ប្រាសាទបន្ទាយស្រី, ខេត្តសៀមរាប}

[2] https://pypi.org/project/khmer-nltk/

IV. Modify the neural language model in III so that the model will learn from scratch the embeddings simultaneously during training. Apply PCA again to these embeddings, plot each word representation on a 2D graph and compare the results with the embeddings produced by the model in I.

Write a report (max 4 pages) summarizing the results from all the implementations.